A one-pass collapse-resistant neural topic model for Mass2Motif discovery

MS2LDA research checkpoint

22 August 2026

Abstract

Mass2Motif discovery requires both predictive topic structure and a broad inventory of chemically interpretable fragment and neutral-loss patterns. We describe a neural probabilistic topic model that amortizes spectrum inference into one sparse top-2 routing pass. The model couples train-only token embeddings, leave-one-out spectrum context, shared topic prototypes, a document-level product-of-experts mixture, mean-normalized fragment/loss evidence, and explicit anti-collapse constraints. We compare the final model with Tomotopy LDA using the same 1,000 topics, seed 42, six CPU threads, frozen scaffold split, and leakage-filtered MAG annotation protocol. On validation, the neural model yields 663 optimized motifs and 185 useful high-confidence motifs, compared with 607 and 138 for Tomotopy. On the once-opened test split, the corresponding counts are 663 and 234 versus 607 and 186. Tomotopy retains higher mean structure overlap score (SOS), exposing a breadth–purity trade-off rather than uniform superiority. The selected neural checkpoint trains in 78.7 minutes versus 76.0 minutes for Tomotopy. This is a fully reproducible single-seed research result, not multi-seed confirmation or a replacement claim for the production backend.

1 Introduction

MS2LDA represents tandem mass spectra as documents, fragment and neutral-loss tokens as words, and recurrent co-occurrence patterns as Mass2Motifs [1]. These motifs can summarize substructures shared across otherwise different molecules. At large scale, useful performance is not a single number: a model should predict held-out spectral evidence, expose many non-redundant motifs, and support chemical annotation of the motifs that receive confident spectrum assignments [2].

Conventional LDA obtains each new spectrum's topic mixture through iterative local inference [7]. A neural alternative can amortize that computation, but large topic inventories are vulnerable to collapse: many prototypes can become unused, duplicate one another, or learn statistically coherent but chemically unhelpful token distributions. A further mass-spectrometry-specific problem is that fragment and neutral-loss vocabularies have different sizes. Summing evidence within each channel therefore rewards the larger vocabulary even when the average token evidence is identical.

This report specifies the final seed-42 neural architecture from input MGF to held-out chemistry. It makes three contributions. First, it defines a one-pass probabilistic router whose document mixture and topic-word decoder share one prototype geometry. Second, it combines complementary anti-collapse mechanisms that act on usage, geometry, and token co-occurrence. Third, it corrects fragment/loss vocabulary-size bias with mean log evidence. The implementation and equations are deliberately presented together so that a new reader can reproduce the model without reconstructing the preceding architecture search.

2 Methods

2.1 Study design and leakage control

The source MGF contains 41,568 spectra. After deterministic cleaning, 38,888 spectra remain. Molecules are keyed by the connectivity block of the InChIKey. Cyclic molecules are grouped by non-chiral Bemis–Murcko scaffold [3]; an acyclic molecule forms its own connectivity group. Whole groups are assigned by descending size to a 70/10/20 train, validation, and test target using seed 42. The resulting matrices contain 27,222 training, 3,889 validation, and 7,777 test spectra, with zero compound or scaffold overlap.

Peak intensities are normalized to the spectrum maximum. Peaks outside $m/z \in [0, 2000]$ or normalized intensity $[0.01, 1]$ are removed; at most 1,000 peaks are kept, and spectra with fewer than five retained peaks are discarded. Each peak creates a fragment token and, when positive, a precursor-minus-peak neutral-loss token, rounded to two decimal places. A token is repeated $\mathrm{round}(100I)$ times, preserving the historical count-based

MS2LDA representation. The vocabulary is built only from training spectra, requires document frequency at least three, preserves first-seen order, and contains 21,233 tokens.

For held-out completion, physical peak groups rather than individual tokens are split deterministically 50/50. A fragment and its neutral loss therefore cannot straddle observed and completion halves. Observed intensities are renormalized without consulting the withheld maximum. All architecture decisions use training and validation only. The selected epoch-40 checkpoint is hashed before the test matrices are opened once.

2.2 Fixed token representation

Let $x_w \in \mathbb{R}^{64}$ denote the fixed feature vector for vocabulary token w. The first 48 dimensions are skip-gram negative-sampling (SGNS) embeddings trained only on training documents [4]. For each of five epochs, eight count-weighted positive token pairs are sampled per document; each pair uses five negatives drawn from corpus frequency raised to 0.75. Source and context embeddings are averaged and normalized.

The token mass m_w contributes 14 Fourier coordinates at frequencies $f \in \{1, 2, 4, 8, 16, 32, 64\}$:

$$\phi_f(m_w) = [\sin(2\pi f m_w/1000)\,, \cos(2\pi f m_w/1000)]\,.$$

The Fourier block is divided by $\sqrt{7}$, and two final one-hot coordinates identify fragment versus neutral loss. Concatenating the normalized 48-D SGNS vector, Fourier block, and type indicators gives 64 dimensions; the complete vector is normalized again. No molecular label or pretrained spectrum encoder enters x_w.

2.3 Shared neural topic geometry and decoder

A learned bias-free projection $W \in \mathbb{R}^{128\times 64}$ maps token features into the unit sphere,

$$z_w = \text{normalize}(W x_w),$$

and each topic has a learned prototype $q_k \in \mathbb{R}^{128}$. The same normalized prototypes define routing and decoding. Topic-word logits are temperature-scaled cosine similarities,

$$\ell_{kw} = \frac{2\,\widehat{q}_k^\top z_w}{\tau_\beta}, \qquad \tau_\beta = 0.18.$$

Let V_F and V_L be the fragment and loss vocabularies. For channel $t \in \{F, L\}$, the model uses log-mean-exp evidence

$$e_{kt} = \text{logsumexp}_{w\in V_t}(\ell_{kw}) - \log|V_t|, \qquad \pi_{kt} = \text{softmax}_t(e_{kt}).$$

Subtracting $\log|V_t|$ is essential: equal average logits now give equal channel evidence regardless of vocabulary size. A fixed $\lambda = 0.25$ pull toward equal fragment/loss mass gives

$$\widetilde{\pi}_{kt} = (1-\lambda)\pi_{kt} + \frac{\lambda}{2}.$$

The row-stochastic decoder is

$$\beta_{kw} = \widetilde{\pi}_{k,t(w)} \frac{\exp(\ell_{kw})}{\sum_{v\in V_{t(w)}} \exp(\ell_{kv})}.$$

This construction preserves rankings within a channel and ensures that the two channel masses sum to one. The final model fixes $\lambda = 0.25$; there is no runtime compatibility switch or alternate decoder in the reproducibility path.

2.4 One-pass spectrum router

For spectrum d, let c_{dw} be its non-zero token count and $s_d = \sum_w c_{dw} z_w$. For an observed token entry w, the leave-one-out context is

$$\bar{z}_{d\setminus w} = \frac{s_d - c_{dw} z_w}{\max(\sum_v c_{dv} - c_{dw}, 1)}.$$

A two-layer MLP with a 256-unit GELU hidden layer and layer normalization produces a residual correction,

$$h_{dw} = \text{normalize}\big(z_w + \text{MLP}([z_w; \bar{z}_{d\setminus w}])\big)\,, \qquad u_d = \text{normalize}(s_d).$$

The unnormalized routing score combines local and whole-spectrum evidence,

$$a_{dwk} = h_{dw}^\top \widehat{q}_k + \rho\, u_d^\top \widehat{q}_k, \qquad \rho = 1.$$

After $r^{\text{soft}}_{dw} = \text{softmax}_k(a_{dwk}/T)$, only the largest two probabilities are retained and renormalized. Training uses the hard-forward, soft-backward straight-through value $r^{\text{hard}} + r^{\text{soft}} - \text{stopgrad}(r^{\text{soft}})$; inference uses the same hard top-2 forward value directly.

Count-weighted routed mass is $M_{dk} = \sum_w c_{dw} r_{dwk}$. A second use of the whole-spectrum evidence sharpens the mixture,

$$g_{dk} = \text{softmax}_k(u_d^\top \widehat{q}_k/T), \qquad \theta_{dk} = \text{normalize}_k(M_{dk} g_{dk}^{\gamma}), \quad \gamma = 0.75.$$

The gate g is detached on this path during training, preventing the model from reducing loss merely by strengthening its own gate; document evidence still learns through a_{dwk}. Multiplication preserves exact zero token support. A spectrum with no routed mass receives the explicit uniform mixture. Thus inference is one projection-and-routing pass, with no per-spectrum variational optimization.

The probabilistic observation model is the ordinary topic mixture

$$p(w \mid d) = \sum_{k=1}^{K} \theta_{dk}\beta_{kw}, \qquad \mathcal{L}_{\text{comp}} = -\frac{1}{\sum_{dw} c_{dw}} \sum_{dw} c_{dw} \log p(w \mid d).$$

The sparse implementation evaluates this expression only at non-zero target tokens; it does not approximate $\theta\beta$.

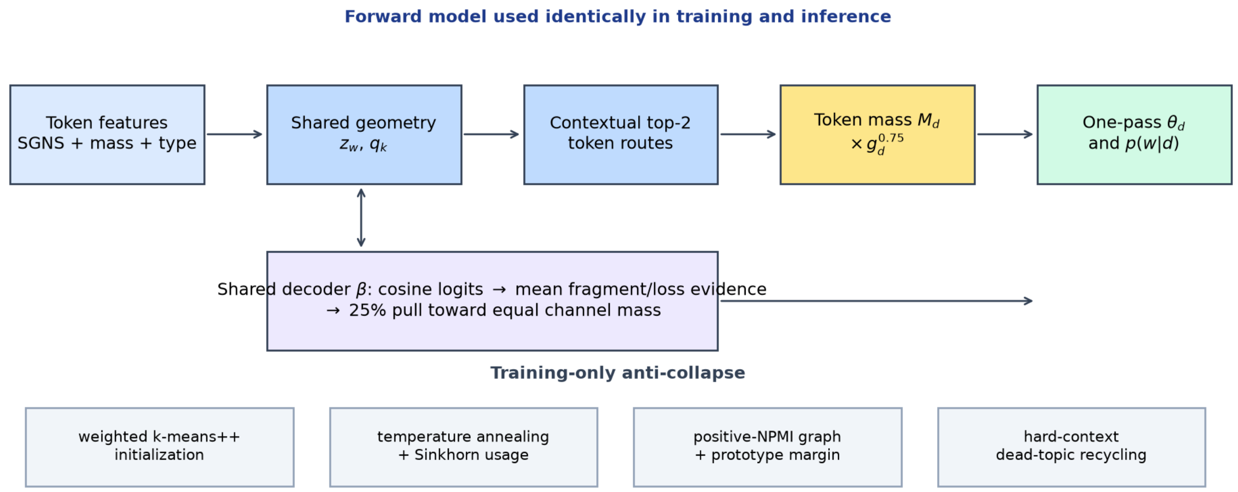

Figure 1: Final model. Fixed train-only token features and learned prototypes define both the one-pass router and the fragment/loss-balanced decoder. The training-only objectives around this forward path prevent collapse.

2.5 Paired-view training objective

Four deterministic view pairs are built from the training MGF. Each side keeps 80 percent of whole physical peak groups (and holds out at least one when possible), then renormalizes intensities independently. Epochs cycle through the four pairs.

Optimization alternates two blocks with one AdamW optimizer. The router block holds β fixed and sends each view through the one-pass router. Its loss is

$$\mathcal{L}_{\text{router}} = \tfrac{1}{2}[\mathcal{L}_{\text{comp}}(\theta^A, X^B) + \mathcal{L}_{\text{comp}}(\theta^B, X^A)] + \omega_S \mathcal{L}_{\text{Sinkhorn}} + 0.1\ \text{JS}(\theta^A, \theta^B) + 5\mathcal{L}_{\text{sep}}.$$

Detached log-space Sinkhorn targets have unit row mass and aggregate topic mass N/K [5, 6]. They regularize only training; they are absent from inference. The topic block freezes the routes and updates prototypes, projection, and β with

$$\mathcal{L}_{\text{topic}} = \mathcal{L}_{\text{comp}} + 0.25\mathcal{L}_{\text{local}} + \mathcal{L}_{\text{NPMI}} + 5\mathcal{L}_{\text{sep}}.$$

The local decoder term asks each selected topic to emit its routed token. Four topic-block updates of batch size 512 follow the full router pass in each epoch.

2.6 Anti-collapse mechanisms

No single regularizer was sufficient during development. The final system uses the following complementary safeguards.

1. **Dispersed initialization.** Prototypes are seeded by deterministic spherical k-means++ without Lloyd updates. Token sampling weights are $\sqrt{\mathrm{CF}_w}\,\mathrm{IDF}_w^2$, balancing common evidence against distinctive words.
2. **Temperature annealing.** Routing temperature decreases linearly from 0.5 to 0.1 over 30 epochs, allowing early exploration before sparse specialization.
3. **Balanced usage.** The Sinkhorn coefficient is 0.25 for the first 10 epochs and then decreases linearly to 0.05 at epoch 40. It resists early winner-take-all collapse without enforcing uniform use at convergence.
4. **Train-only positive-NPMI geometry.** A binary-document graph keeps tokens with document frequency at least 10, pair frequency at least three, positive NPMI, and at most 16 strongest neighbours. Only mutual neighbours remain. With sparse adjacency A, the loss is

$$\mathcal{L}_{\mathrm{NPMI}} = -\frac{1}{K}\sum_k \log\left(\sum_{wv} \beta_{kw} A_{wv} \beta_{kv}\right).$$

   It encourages a topic's probability mass to lie on training-supported token pairs rather than isolated high-logit words.
5. **Prototype separation.** For each prototype, the eight nearest other prototypes are penalized above cosine margin 0.3:

$$\mathcal{L}_{\mathrm{sep}} = \mathrm{mean}_{k,j}[\max(0, \widehat{q}_k^\top \widehat{q}_j - 0.3)]^2.$$

   This addresses geometric duplication, which balanced usage alone cannot see.
6. **Deterministic dead-topic recycling.** At every second epoch, a topic below $0.1/K$ corpus usage for three consecutive validations is replaced by a stored high-loss routing context. Adam state for that row is cleared. Recycling is allowed through epoch 40 and at most twice per topic; 28 replacements occurred in the selected run.
7. **Numerical controls.** All losses and gradients must be finite; gradient norm is clipped to 5.0. Selection is the predeclared final epoch 40, not the best validation checkpoint.

Training uses AdamW with learning rate 10^{-3}, weight decay 10^{-4}, router batch size 128, and seed 42. Table 3 in the appendix is generated directly from the committed protocol.

2.7 Tomotopy comparator

The comparator is trained independently from the same frozen training matrix; it never acts as teacher. Tomotopy 0.13.0 [8] uses $K = 1000$, scalar initial $\alpha = 0.6$ with optimization interval 10, $\eta = 0.1$, seed 42, six workers, and parallel scheme 3. Training proceeds in 50-iteration blocks to at most 5,000 iterations and stops when each of the last ten relative perplexity changes is below 0.005; the recorded model stopped after 750 iterations. These settings match the benchmark notebook except that its `workers=0` (all available cores) is replaced by the shared six-worker cap for both methods. Held-out mixtures use 100 inference iterations, six workers, parallel scheme 1, and independent-document inference. The neural model has no analogue of this iteration count because its held-out mapping is amortized into one forward pass.

2.8 Evaluation and chemical annotation

The predictive metric is count-weighted held-out completion negative log-likelihood (NLL) per in-vocabulary token. Chemical evaluation follows the paper-aligned MAG workflow [2]. Before retrieval, every library entry sharing a validation or test connectivity key is removed. For each topic, the 20 highest-probability tokens form a motif spectrum. The search retrieves up to 1,000 neighbours, retains ten unique molecules, clusters at cosine similarity 0.9, and optimizes motif features. A motif with at least one optimized feature counts toward MAG annotation coverage.

High-confidence topic–spectrum associations require $\theta_{dk} \geq 0.5$. SOS is computed against unique compounds, so repeated spectra do not overweight a molecule. Among SOS-evaluable high-confidence motifs, the fixed bands are high (> 0.8), intermediate ($0.6 \leq \mathrm{SOS} \leq 0.8$), and low ($< 0.6$). A *useful high-confidence motif* is in the high or intermediate band. These inventory quantities, mean/median SOS, and model-fitting time are the primary comparison outputs. Held-out NLL is retained as secondary predictive evidence alongside collapse diagnostics and warm inference in the appendix.

3 Results

3.1 Validation comparison

Table 1: Primary validation comparison.

Metric	Neural	Tomotopy
Optimized motifs	663	607
MAG annotation coverage	66.3%	60.7%
High-confidence SOS-evaluable motifs	312	206
SOS high (> 0.8)	49	58
SOS intermediate (0.6–0.8)	136	80
SOS low (< 0.6)	127	68
Useful high-confidence motifs (SOS $\geq$ 0.6)	185	138
Mean high-confidence SOS	0.6323	0.6761
Median high-confidence SOS	0.6364	0.6854
Model-fitting time (minutes)	78.7	76.0

The final neural model produces 663 optimized motifs, 56 more than Tomotopy, and 312 high-confidence SOS-evaluable motifs, 106 more than Tomotopy. Its 185 useful motifs exceed Tomotopy's 138 by 34.1 percent. The price of this broader inventory is lower mean SOS (0.632 versus 0.676), although the neural mean remains above the predeclared 0.6318 floor. Model fitting is similar: 78.7 minutes for neural and 76.0 minutes for Tomotopy.

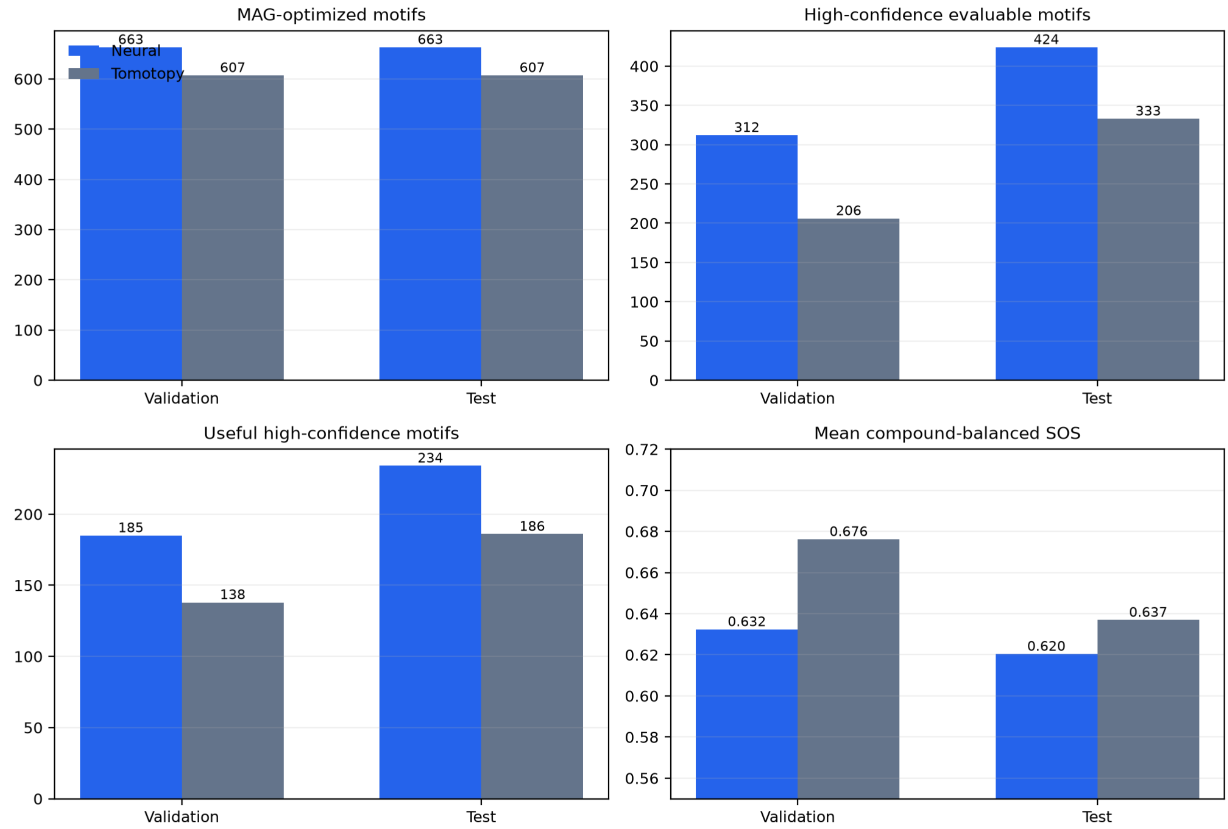

Figure 2: Paper-relevant validation and test outcomes. Counts favour the neural model, while mean SOS exposes the remaining quality trade-off.

3.2 Locked test confirmation

Table 2: One-time test comparison after checkpoint locking.

Metric	Neural	Tomotopy
Optimized motifs	663	607
MAG annotation coverage	66.3%	60.7%
High-confidence SOS-evaluable motifs	424	333
SOS high (> 0.8)	58	66
SOS intermediate (0.6–0.8)	176	120
SOS low (< 0.6)	190	147
Useful high-confidence motifs (SOS ≥ 0.6)	234	186
Mean high-confidence SOS	0.6204	0.6369
Median high-confidence SOS	0.6135	0.6145
Model-fitting time (minutes)	78.7	76.0

The test split confirms the main pattern. Neural MS2LDA yields 234 useful high-confidence motifs versus 186 for Tomotopy and 424 SOS-evaluable motifs versus 333. Coverage remains 66.3 versus 60.7 percent. Test mean SOS is lower (0.620 versus 0.637), whereas medians are almost identical (0.613 versus 0.615).

4 Discussion

The final result suggests that the last annotation-coverage failure was partly an accounting bias in the decoder, not a need for a larger network. Log-sum-exp channel evidence includes a $\log |V_t|$ multiplicity term; removing it made fragment and neutral-loss evidence comparable and allowed the existing soft balance to operate as intended. Because this correction adds no parameter, its gain is likely to be more portable than a dataset-specific capacity increase.

Parity is therefore metric-dependent. On the paper-facing inventory metrics, the neural model reaches or exceeds practical parity: it produces more optimized, evaluable, and useful motifs and fits held-out spectral evidence better at similar training time. Tomotopy still has higher average validation and test SOS. The neural model is broader, not uniformly purer. The result also uses one seed by design. A large effect under seed 42 supports the architecture, but it does not quantify seed uncertainty.

The most consequential future extension is a jointly trained spectrum-level encoder. The present u_d is a normalized sum of token embeddings; a learned set or transformer encoder could capture global peak relations that no sum can represent, while the current prototype decoder would preserve interpretable motifs. This has greater potential impact than selective motif freezing, which was already feasible in variational-Bayes LDA. Other documented extensions are:

- sparse chemical supervision or structural constraints only for spectra where trustworthy labels exist, retaining the unsupervised objective elsewhere;
- fine-tuning routing and motif representations on a new dataset while monitoring vocabulary drift and collapse;
- GPU-native sparse batching for much larger corpora and encoders; and
- optionally freezing selected known prototypes while allowing the remaining topics to learn, useful operationally but not conceptually unique to the neural formulation.

The neural formulation makes these extensions differentiable and composable; none is implemented or included in the present result.

Limitations are deliberate and concrete. This is a single-dataset, single-seed comparison. Warm inference excludes file loading, preprocessing, and first-call setup. MAG annotation depends on the retrieval library and optimization procedure. Finally, topic counts and SOS characterize the evaluable inventory rather than proving that every motif is a distinct biochemical substructure.

5 Conclusion

A compact combination of one-pass amortized routing, shared topic geometry, mean-normalized fragment/loss evidence, and explicit anti-collapse training produces a broad neural Mass2Motif inventory without sacrificing predictive fit. Under the fixed seed-42, $K = 1000$, six-thread protocol, the model exceeds Tomotopy in optimized and useful motif counts, while Tomotopy retains higher mean SOS. The code, protocol, model bundle, generated tables, and checkpoint hash form a shareable reproducibility checkpoint. The appropriate next scientific step is replication, not further tuning on the locked test split.

A Exact hyperparameters

Table 3: Values generated from `protocol.json`.

Quantity	Value
Seed; topics; CPU threads	42; 1000; 6
Token features	48 SGNS + 14 Fourier + 2 type = 64 dimensions
SGNS	5 epochs; 8 pairs/document; 5 negatives; power 0.75; lr 0.01
Projection; router hidden	128; 256
Decoder	temperature 0.18; type pull 0.25
Router	top-2; additive document evidence; gate exponent 0.75
Views	4 pairs; 80% physical peak groups/view
AdamW	lr 0.001; weight decay 0.0001; gradient clip 5.0
Batches and epochs	router 128; topic 512; 4 topic updates/epoch; 40 epochs
Loss weights	local 0.25; view consistency 0.1; NPMI 1.0; separation 5.0
Routing temperature	0.5 to 0.1 over 30 epochs
Sinkhorn	epsilon 0.05; 5 iterations; weight 0.25 to 0.05
NPMI graph	min DF 10; min pair 3; 16 neighbours; min NPMI 0.0
Prototype separation	8 neighbours; margin 0.3
Recycling	usage $<$ 0.1 of uniform for 3 validations; max 2/topic
Tomotopy	alpha 0.6; eta 0.1; step 50; max 5000; inference 100
MAG/SOS	top 20 motif tokens; membership $\geq$ 0.5; cluster cosine 0.9

B Equation–implementation correspondence

Table 4: Direct map from this report to executable code.

Report component	Implementation
Token feature x_w	`data.py: build_token_features`
Projected token z_w	`model.py: projected_tokens`
Decoder β and mean type evidence	`model.py: topic_word_distribution`
Leave-one-out h_{dw} and u_d	`model.py: _route_embeddings`
Top-2 r_{dwk}	`model.py: route`
Gated mixture θ_d	`model.py: aggregate_theta`
Completion likelihood	`model.py: sparse_completion_nll`
Router and topic objectives	`objectives.py`
Positive-NPMI graph	`cooccurrence.py`
NPMI and separation losses	`objectives.py`
Alternating optimization and recycling	`optimization.py; training.py`
One-pass held-out inference	`model.py: infer_theta`
Tomotopy fit and inference	`tomotopy.py`
MAG/SOS evaluation	`chemical.py`

C Secondary diagnostics and practical inference

At test time, 319 neural topics exceed uniform corpus usage, and median effective topics per spectrum are 7.28. These are collapse-safety diagnostics, not headline chemical outcomes. Held-out completion NLL per token was 8.5014 versus 9.6622 on validation and 8.5226 versus 9.7569 on test (neural versus Tomotopy). This predictive check is reported here as safety evidence rather than as a headline chemistry result.

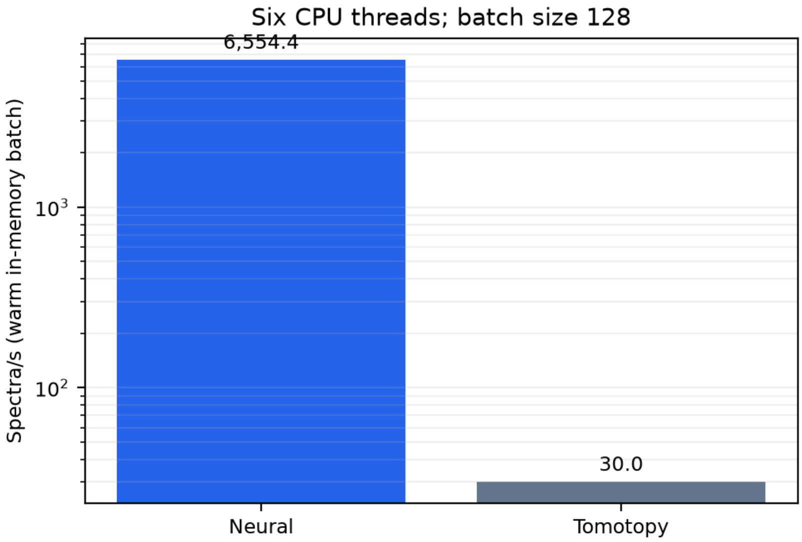

Figure 3: Matched six-thread warm in-memory batch inference after one warm-up. The neural model uses one routing pass; Tomotopy uses 100 inference iterations. This is not end-to-end application latency.

The accepted checkpoint processes 6,554 spectra/s in a resident 128-spectrum batch versus 30.0 spectra/s for Tomotopy, a 218-fold ratio. This is usable for repeated in-memory batches or a resident service, but excludes parsing, feature construction, model loading, and cold-start costs.

D Reproducibility record

The protocol fixes seed 42, $K = 1000$, six CPU threads, and final epoch 40. The portable bundle contains the resolved protocol, vocabulary, model state, provenance, and file hashes; evaluation evidence remains in the canonical `results.json` outside the bundle. The selected checkpoint SHA-256 is:

`639c1f37c613d908b59e3a85b7dc701e`
`33a3f92fd7476a3257b47298143dfbc6`.

From the repository root, a clean reproduction uses the checked-in neural and MAG environments, then downloads and verifies Zenodo record 20179680:

```
conda env create -f environment-neural-ms2lda.yml
conda env create -f environment-msnlib-mag.yml
conda run -n ms2lda-neural python \
  scripts/download_msnlib_validation_assets.py \
  --data-root /path/to/MSnLib-assets
conda run -n ms2lda-neural python -m benchmarks.neural_ms2lda run \
  --data-root /path/to/MSnLib-assets --run /path/to/run
conda run -n ms2lda-neural python -m benchmarks.neural_ms2lda verify \
  --data-root /path/to/MSnLib-assets --run /path/to/run
```

The acquisition script checks archive and extracted-file hashes before the run begins. The runner records source, protocol, dependency, and input hashes; the same `run` command resumes exactly at a completed epoch or stage boundary. The workflow trains the Tomotopy comparator from the same frozen training matrix; no external reference run is required. Validation evaluation and MAG scoring complete before the test evaluation functions are called.

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